Arsenii Harbar

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MInf Computer Science & AI, University of Edinburgh | Applied AI & ML Systems — three first-place hackathon finishes, two at Google-partnered events

Education

University of Edinburgh Edinburgh, UK
*MInf (Integrated Masters) Computer Science & Artificial Intelligence — **undergraduate, UG4*** *Expected May 2028*

- Began university at **16**, First Class
- **AI & systems coursework:** Machine Learning Systems, High-Performance Computing, Parallel Programming Languages & Systems, Computer Vision (**MSc-level**), Financial Markets Applications of Deep & Reinforcement Learning **(PhD Level)**.
- **Final-year project:** continuing the multi-objective optimisation work with the Large-Scale Machine Learning Systems group.

Hackathons & Competition Wins

- **1st Place — NatWest × Google NextGenHack** (2026). Winning **four-person** team, competing as a third-year undergraduate in a predominantly **postgraduate** field. Designed, built and demoed on **Google Cloud** inside the event window.
- **1st of 40+ teams — Skyscanner × Google AccessAIthon** (2026). Accessibility-focused AI build delivered under a single-day deadline, from problem framing through to a working demo and pitch.
- **1st Place — AI Expo**, Edinburgh AI × Anthropic. Judged on the applied AI build and its presentation.
- **Certifications: GPU Programming** (Johns Hopkins, **93%**) — capstone was GPU-accelerated speech recognition in CUDA/PyTorch; **Deep Learning Specialization** (DeepLearning.AI, **95%**).

Research & Industry Experience

Research Assistant — Multi-Objective Optimisation for LLM Training Jun. – Aug. 2026
Large-Scale Machine Learning Systems Group — Dr Luo Mai *Edinburgh, UK*

- Worked on **Jacobian-descent methods for multi-objective optimisation** of transformer models, within a group researching how large models are trained efficiently.
- Built the **profiling and benchmarking infrastructure** used to evaluate the approach across model and batch scales, running repeatable measurement campaigns on **SLURM GPU clusters** with per-phase memory and wall-clock capture.
- **Stack:** PyTorch 2.4.1 / CUDA 12.1, HuggingFace, nanoGPT, GPT-2 124M and Qwen, NVIDIA A5000, SLURM, `nsys` profiling.

Undergraduate Research Assistant — Data Systems & NLP Jan. 2025 – Jun. 2026
University of Edinburgh — Prof. Mustapha Douch *Edinburgh, UK*

- Built high-throughput ingestion pipelines over **200M+** unstructured records under tight memory limits; **3**× throughput via vectorised tokenisation and chunked streaming, halving RAM footprint.
- Trained and deployed ensemble classifiers at **97%+** accuracy, **+12 points** over the prior production baseline, validated before rollout.

Projects & Leadership

RiskBeacon — Real-Time Market Risk Monitoring | *Python, FinBERT, FastAPI, Streamlit, MLflow, Docker* | GitHub Nov. 2025 – Present

- End-to-end system ingesting OHLCV market data and computing quantitative stress metrics — **VPIN** (volume-synchronised probability of informed trading), **Yang–Zhang volatility** and market-regime classification — under a **sub-100 ms** end-to-end latency budget.
- Service-oriented build: **FastAPI** REST backend, **Streamlit**/Plotly dashboard, **FinBERT** news-sentiment service for market context, and **MLflow** tracking for model reproducibility, containerised for deployment.
- Treated as production software, not a demo: **200+ property-based tests** (Hypothesis) enforcing runtime invariants, plus integration tests, **Great Expectations** data-validation suites and strategy backtests, all on GitHub Actions CI.

Face Detection & Anonymisation Pipeline | *C++, Python, OpenCV* | GitHub Aug. 2025

- Two-stage batch pipeline: a Python/OpenCV Haar-cascade stage detects faces across an image corpus and emits bounding boxes, which a **C++** stage consumes to apply Gaussian blur over every detected region — splitting the work so the per-image processing runs in compiled code.

ML Development & Research Lead Sep. 2024 – Present
Algorithmic Trading & Quantitative Finance Society (QuantSoc), University of Edinburgh *Edinburgh, UK*

- Lead the ML and optimisation workstream for a **1,000+ member** society: strategy research and large-scale backtesting orchestrated on university GPU compute, covering experiment scheduling, result capture and comparison at scale.
- Design quantitative problems and deliver live workshops on signal generation and execution algorithms to **100+** students; secured **Hudson River Trading** as the society's first corporate sponsor alongside two co-leads, and run an active pipeline of quantitative-firm and university partnerships.

Technical Skills

AI & Machine Learning: PyTorch, HuggingFace Transformers, LLM APIs and prompt engineering, FinBERT, NumPy/Pandas, XGBoost, PPO/MARL
Cloud & Delivery: Google Cloud, Docker, CI/CD (GitHub Actions), SLURM GPU clusters, property-based testing (Hypothesis)
Languages: Python, C/C++ (CUDA), Java, SQL, Haskell, Bash
Systems & Performance: custom CUDA kernels, `nsys` profiling, memory/throughput benchmarking, SLURM GPU clusters
Data & Serving: FastAPI, Streamlit, MLflow, Great Expectations, OpenCV, Plotly, pandas/NumPy